

SdCvP-X™: Electronic (Design) Verification Automation Tool

VERIPOINT TECHNOLOGIES PVT LTD

AI Tools & Solutions Powering Semiconductors and Beyond

Company Profile

Veripoint Technologies is a deep-tech product company addressing critical industry challenges, from automating semiconductor verification to delivering scalable, intelligent automation for industrial sectors through its AI-powered tools and engineering solutions.

Industry Challenges

- **Semiconductors:** Manual verification slows development and increases the risk of costly silicon failures.
- **Industrial Sectors:** Scalable, intelligent automation solutions are still limited, hindering operations, efficiency and innovation.

Solutions: Our Flagship Products & Premium DV Services

- **SdCvP-X™** : Proprietary AI-driven Electronic (Design) Verification Automation Tool.
- **AI-driven solutions** for manufacturing, automotive, telecommunications, healthcare and more.
- **Premium DV services** delivered by a team of experts with deep specialization in verifying complex designs.

Founded: Feb 2022 | **HQ:** Noida, UP, India | **Website:** www.veripointtech.com | **DPIIT:** DIPP158213

SdCvP-X™: Electronic (Design) Verification Automation Tool

SdCvP-X™ is our proprietary AI tool that automates chip design verification, accelerating development and ensuring thorough verification.

Target Designs

- IP (standard & custom), SoC, and Subsystem

Specification (Design)

Benchmarks

- High-speed and standard Interfaces like PCIe, CXL, NVMe, Ethernet, UCIe, Display, ARINC
- Custom, Automotive & Networking chip interfaces
- Custom Analog IPs & Systems

Key Features

- **Auto-generates** features, tests, assertions, and coverage plans from specifications in various formats including **IP-XACT**
- **Auto-generates** UVM test sequences and SystemVerilog assertions with custom format options
- **Dynamically adapts** to evolving specifications and syncs with manual plans
- **Auto-embeds** debug data for faster root-cause analysis
- **Integrates** with existing verification flows via Vplan for leading EDA tools

SdCvP-X™: Verifying Complex Designs

Verification Challenges	SdCvP-X™ Tool
Manual & fragmented planning, tests and assertions gaps	Auto-generates spec-aligned verification plans, UVM test sequences, and SystemVerilog assertions for Digital (IP, SoC, Subsystem) and Analog designs, directly from specifications including IP-XACT
Specification Alignment	Plan evolves with specification + Version traceability
Manual Plan Alignment with Specification and upgrades	Aligns manual plans to eliminate gaps and ensure consistency and boost productivity.
Debug Inefficiencies	Traceability matrix + mapping to features/test points
Configuration Explosion	Smart filtering, collapsing, and prioritization
Reuse across derivatives	Scalable planning across standard & custom designs

Who Benefits: Across the Project Hierarchy

Role	Pain Point	SdCvP-X™ Benefit
Engineers (DV)	Long debug cycles, manual test list creation	Faster test planning + comprehensive tests and assertions + structured debug : Boosts Productivity
Architects	Hard to validate spec implementation	Full traceability from spec to test
Program/Project Managers	Poor visibility on verification progress	Actionable insights, milestone-based views
Business Leaders & Industries	Risk of re-spin, time-to-market delays	Higher First Silicon Success, faster Time to market

SdCvP-X™: GUI & OUTPUT for CXL 3.1 IP (Eval) Interface

Design Domain

☒ IP ☐ SoC ☐ Subsystem

Upload Specification

Base Specification (PDF/docs)

CXL-3.1-Specification © 2019-2023 COMPUTE EXPRESS LINK CONSORTIUM, INC. ALL RIGHTS RESERVED.pdf

Base Register Specification (.xlsx/PDF)

Base Add-on Specification (PDF/docs)

Base Add-on Register Specification (.xlsx/PDF)

Feature List

Do you want specification feature list extraction?

☒ Yes ☐ No

Download Feature List

Do you want to upload modified feature list according to your design?

☐ Yes ☒ No

Verification Plan Filters

Configuration Enabled ☒ Yes ☐ No

Constraints Enabled ☒ Yes ☐ No

Coverage Enabled ☒ Yes ☐ No

Assertion Enabled ☒ Yes ☐ No

Debug Enabled ☒ Yes ☐ No

UVM Test Sequence Enabled ☒ Yes ☐ No

Confirm Filters

Generate Verification Plan

transcript_cxl_3_1.txt

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Domain selected: IP

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Reading Specification: CXL-3.1-Specification © 2019-2023 COMPUTE EXPRESS LINK CONSORTIUM, INC. ALL RIGHTS RESERVED.pdf

Created Processed Data: TEMP_Data_SdCvP-x from Specification CXL-3.1-Specification.pdf

Extracted 16 features.

Feature list saved : /Users/Veripoint_Technologies/Desktop/sdcvp-x/feature_list_cxl_3_1.xlsx

Verification Plan Filters confirmed:

Configuration Enabled: Yes

Constraints Enabled: Yes

Coverage Enabled: Yes

Assertion Enabled: Yes

Debug Enabled: Yes

UVM Test Sequence Enabled: Yes

Verification plan generated and saved: /Users/Veripoint_Technologies/Desktop/sdcvp-x/verification_plan_cxl_3_1.xlsx

Deleted Processed Data: TEMP_Data_SdCvP-x Securing Design Confidentiality

Verification Plan			Simulation Output									
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S.No.	Section	Layer/Component	Feature	DUT Type	Testpoint	Test Scenario	Procedure	Constraint	Assertion	Coverage	Spec Variables	Register Variables
Verification Plan			Simulation Output									

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S.No.	Feature	Test Scenario	UVM Test Sequence				System Verilog Assertions				

SdCvP-X™: GUI & OUTPUT for Analog PM Bus Interface

☒ SoC
 ☐ Subsystem

Upload Specification

Base Specification (PDF/docs)

Topic_6_Hesse.pdf

Base Register Specification (.xlsx/PDF)

Base Add-on Specification (PDF/docs)

Base Add-on Register Specification (.xlsx/PDF)

Feature List

Do you want specification feature list extraction?

Yes No

Download Feature List

Do you want to upload modified feature list according to your design?

Yes No

Verification Plan Filters

Configuration Enabled

Constraints Enabled

Coverage Enabled

Assertion Enabled

Debug Enabled

UVM Test Sequence Enabled

Yes No

Yes No

Yes No

Yes No

Yes No

Yes No

Confirm Filters

Generate Verification Plan

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transcript_pm_bus.txt

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Domain selected: IP
Uploaded: Topic_6_Hesse.pdf
Reading Specification: Topic_6_Hesse.pdf
Created Processed Data: TEMP_Data_SdCvP-x from Specification Topic_6_Hesse.pdf
Extracted 33 features.
Feature list saved : /Users/Veripoint_Technologies/Desktop/sdcvp-x/feature_list_pm_bus.xlsx
Verification Plan Filters confirmed:
Configuration Enabled: Yes
Constraints Enabled: Yes
Coverage Enabled: Yes
Assertion Enabled: Yes
Debug Enabled: Yes
UVM Test Sequence Enabled: Yes
Verification plan generated and saved: /Users/Veripoint_Technologies/Desktop/sdcvp-x/verification_plan_pm_bus.xlsx
Deleted Processed Data: TEMP_Data_SdCvP-x Securing Design Confidentiality
  
```

Verification Plan

Simulation Output

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S.No.	Section	Layer/Component	Feature	DUT Type	Testpoint	Test Scenario	Procedure	Constraint	Assertion	Coverage	Spec Variables	Register Variables
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Verification Plan

Simulation Output

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S.No.	Feature	Test Scenario	UVM Test Sequence	System Verilog Assertions
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SdCvP-X™: Tiered Product Options

Standard

- ✓ Comprehensive Verification Plan
- ✓ Features, Test Points & Scenarios
- ✓ Configuration & Constraints Plan

Advanced

- ✓ All Standard Features
- ✓ Assertion Plan
- ✓ Enhanced Specification Evolution
- ✓ Testplan Alignment

Most Popular

Premium

- ✓ All Advanced Features
- ✓ Coverage Plan
- ✓ Auto-Generated UVM Test Sequences and System Verilog Assertions
- ✓ Spec Variables, Register Variables, Debug Info & Traceability
- ✓ Priority Support & Success Manager

Contact Us

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